FILTER DESIGN TEST CASE

High-Pass Filter - Band 2 (1900 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0007
Revision:	Rev D.6
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Created Date:	2024-03-21
Last Modified:	2024-05-06
Status:	CONDITIONAL PASS
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 2 (1900 MHz) for Public Safety Band Filter. The filter is designed on 36° YX $LiTaO_3$ substrate with a target center frequency of 5244.8 MHz and bandwidth of 280.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 2 (1900 MHz)
- Application: Public Safety Band Filter
- Substrate: 36° YX $LiTaO_3$
- Package: WLP (Wafer Level Package)
- Design Tool: Cadence AWR Microwave Office
- Design Center: Newbury Park Design Center

This specification applies to all variants of the High-Pass Filter design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 5244.8 MHz
Bandwidth (3 dB): 280.3 MHz
Insertion Loss Target: <= 2.1 dB
Return Loss Target: >= 20.0 dB
Out-of-Band Rejection: >= 28.0 dB
Isolation (Duplexer): >= 36.0 dB
Q Factor: 7969
Temperature Coefficient:-11.4 ppm/°C
Die Size: 1.8 x 1.0 mm
Wafer Size: 8-inch
Number of Resonators: 8
Electrode Material: Pt/Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the High-Pass Filter template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 5244.8 MHz, BW 280.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15734 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.1 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 420 MHz offset >= 28.0 dB
- Group delay variation across passband shall be <= 9 ns
- Temperature drift over -40°C to +85°C shall be <= 7.5 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-05-06
Devices Tested: 45
Insertion Loss (meas): 1.83 dB
Return Loss (meas): 21.8 dB
Rejection (meas): 30.8 dB
Isolation (meas): 41.1 dB
Q Factor (meas): 7969
Group Delay Variation: 3.7 ns
Power Handling: +32 dBm
Die Yield: 90.7%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

a) The High-Pass Filter design demonstrated acceptable performance for Band 2 (1900 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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